

Spike Growth on Patterned Gold Nanoparticle Scaffolds

Jin Jia, Nadia Metzkow, Sang-Min Park, Yuhao Leo Wu, Alexander D. Sample, Bundit Diloknawarit, Insub Jung, and Teri W. Odom*



Cite This: *Nano Lett.* 2023, 23, 11260–11265



Read Online

ACCESS |



Metrics & More



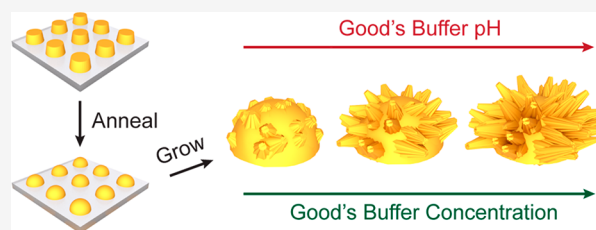
Article Recommendations



Supporting Information

ABSTRACT: This work reports a scaffold-templated, bottom-up synthesis of 3D anisotropic nanostructures on periodic arrays of gold nanoparticles (AuNPs). Our method relies on substrate-bound AuNPs as large seeds with hemispherical shapes and smooth surfaces after the thermal annealing of as-fabricated particles. Spiky features were grown by immersing the patterned AuNPs into a growth solution consisting of a gold salt and Good's buffer; the number and length of spikes could be tuned by changing the solution pH and buffer concentration. Intermediate structures that informed the growth mechanism were characterized as a function of time by correlating the optical properties and spike features. Large-area (cm^2) spiky AuNP arrays exhibited surface-enhanced Raman spectroscopy enhancement that was associated with increased numbers of high-aspect-ratio spikes formed on the AuNP seeds.

KEYWORDS: spiky gold nanoparticle array, seed-mediated growth, nanoparticle growth, nanofabrication, surface-enhanced Raman spectroscopy



Controlling the nanoscale surface features on anisotropic gold nanoparticles (AuNPs) opens prospects to realize architectures with exquisitely tunable optical properties.^{1–4} AuNPs constructed from a central core and protruding spikes—spiky AuNPs—are a class of anisotropic AuNPs that exhibit multiple localized surface plasmon (LSP) resonances with different resonant wavelengths and strong electromagnetic fields near the sharp tips.⁵ Such properties are critical in photoacoustic imaging,⁶ photothermal applications,⁷ and surface-enhanced Raman spectroscopy (SERS).⁸ Previous work has shown that the crystallinity and shape of intermediate structures can determine the final morphology of anisotropic AuNPs such as gold nanostars grown in a seedless synthesis;^{9,10} however, due to room-temperature growth conditions, these particles are highly heterogeneous in size and shape.¹¹ Typically, seed-mediated methods are used to synthesize spiky AuNPs with narrower size and shape distributions, where small (<30 nm) seeds are introduced into a growth solution containing gold salts and shape-directing agents, such as ascorbic acid,¹² cetyltrimethylammonium bromide,¹³ and polyvinylpyrrolidone.¹⁴ Visualizing the reaction intermediates during the growth of seeded spiky AuNPs would provide insight into the synthetic mechanism and a tool to manipulate the final NP shape.

Recently, larger (30–100 nm) seeds have been used to prepare spiky AuNPs in a Good's buffer-based system.¹⁵ Spiky AuNPs synthesized from small seeds showed much larger spikes compared to their core sizes, while NPs synthesized starting from large seeds had shorter but higher numbers of spikes.¹⁵ Larger spiky NPs exhibit stronger scattering properties and enhanced electromagnetic near-fields from their increased

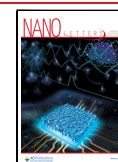
numbers of sharp spikes.^{14,16} Drawbacks to synthesize spikes on colloidal seeds via solution with diameters greater than 100 nm include lack of uniform structures due to the heterogeneous sizes and shapes of large AuNPs and instability in solvents.¹⁷ Although top-down methods, including nanoimprint lithography,¹⁸ nanosphere lithography,¹⁹ electron-beam lithography,²⁰ and photolithography²¹ are deterministic approaches to pattern NPs,²² fabrication of features at sub-10 nm levels is challenging because of resolution limitations and in maintaining dimensional accuracy during multistep lithographic processes.²³ However, there could be benefits for using patterned NPs as large seeds to synthesize anisotropic nanostructures: (1) the surfaces do not require ligands for stabilization;²⁴ (2) patterned scaffolds (NP diameter $d > 100$ nm) can be easily created;²⁵ and (3) final sizes and shapes can be very uniform.^{25,26} For example, nanoimprint lithography has been used to fabricate substrate-bound AuNP seeds for templated synthesis of faceted Au nanostructures²⁷ and nanostars.²⁸ A general approach combining NP scaffolds produced by top-down methods and nanoscale feature growth by bottom-up syntheses that could be observed as a function of time would enable the design of anisotropic NPs with multiscale control.

Received: October 3, 2023

Revised: November 22, 2023

Accepted: November 28, 2023

Published: December 4, 2023



Here we demonstrate a scaffold-guided synthesis to grow spikes on periodic arrays of AuNPs. Uniform hemispherical AuNPs ($d = 70\text{--}190\text{ nm}$) with smooth surfaces functioned as patterned seeds that were immersed in a growth solution of a gold salt (HAuCl_4) and Good's buffer (EPPS). Based on conditions for spiky NPs from colloidal seeds, solution pH and EPPS concentration were used to manipulate the spike and NP core dimensions. At pH 7.5 ($< \text{EPPS } pK_a, 8.0$), high numbers and long lengths of spikes were observed, while at lower pH (6.0), no spikes were formed. When the EPPS concentration was 200 mM, we observed the initiation of sharp spikes; at 500 mM, both the number and length started to decrease. By correlating the reaction time with structural and optical properties of intermediate species during spike growth, we found that AuNP seeds first developed facets at shorter reaction times (1 min), and then small, short protrusions formed, which gradually evolved into sharp, longer spikes after ca. 30 min. Exquisite control over the seed size and growth conditions has resulted in large-area spiky AuNP arrays with uniform structures, tunable optical responses, and strong SERS enhancement.

Figure 1a depicts the scheme to synthesize spikes on the periodic arrays of AuNPs. Patterned AuNPs ($a_0 = 600\text{ nm}$, $d = 70\text{ nm}$) on quartz substrates were fabricated by SANE (solvent-

roughness of the pattern transfer masks in the physical vapor deposition process, as-fabricated AuNPs were polycrystalline and had irregular cylindrical shapes (Figure 1b and Figure S2a). To form spikes on the patterned AuNPs, the substrate was immersed in a growth solution consisting of HAuCl_4 (0.4 mM) and EPPS (400 mM) at pH 7.5 for 16 h at room temperature. Spiky AuNPs synthesized from as-fabricated seeds had irregular shapes, variations in NP size, and a heterogeneous distribution of spike numbers and lengths (Figure 1c). To improve the uniformity and surface crystallinity of the AuNPs, we annealed the as-fabricated seeds at $800\text{ }^\circ\text{C}$ under a pressure of 1.2 Torr in Ar for 30 min to form hemispherical shapes with smooth surfaces (Figure 1d and Figure S2b). Under the same growth conditions, spiky AuNPs formed from the annealed AuNP seeds showed a high degree of structural uniformity in both the central core size and the spike dimensions (Figure 1e). Similar to seed-mediated growth in solution,^{17,31,32} the shape and surface crystallinity of the seeds are important for anisotropic NP growth.

Solution pH is a key chemical factor that affects the formation of spiky features.³³ Using annealed patterned AuNPs as seeds and fixed concentrations of EPPS (400 mM) and HAuCl_4 (0.4 mM), we tested growth solutions at different pH values (6.0, 6.5, 7.0, and 7.5). We kept the pH of all solutions below the pK_a (8.0) of EPPS since AuNPs synthesized in solution with $\text{pH} > \text{buffer } pK_a$ are not stable.³⁴ At pH 7.5, we observed high numbers and long lengths of spikes, while at pH 6.0, no spike formation was observed (Figure 2a–d). For all pH conditions, the synthesized AuNPs showed near-spherical central core structures that were larger ($>100\text{ nm}$) than the initial seeds ($d = 70\text{ nm}$) (Figure 2e). To compare differences in spike numbers in spiky NPs, we defined a comparative metric, normalized peripheral spike number, to represent the overall degree of branching. Only peripheral spikes outside of the central cores were counted (Figure 2f, green ring), and then this number was normalized to the highest spike number from AuNPs synthesized with EPPS (400 mM) at pH 7.5. For EPPS at pH 6.0, the synthesized AuNPs had no spikes and showed only one LSP resonance at 660 nm (Figure 2g). At pH 6.5, a slight red shift in the LSP resonance wavelength (at 680 nm) was observed because of small protrusions on the AuNPs. Distinct short spikes were present starting from pH 7.0 (Figure 2h), and a second LSP peak appeared at 900 nm with a low intensity. As pH increased to 7.5, the spiky AuNPs showed two strong LSP resonances at wavelengths of 700 and 900 nm from the higher numbers of low- and high-aspect-ratio spikes.

After establishing pH 7.5 as optimal for growing large numbers of long spikes, we examined the effect of EPPS concentration on spike growth, keeping the gold salt concentration fixed (0.4 mM). EPPS buffer produced spikes throughout the entire range of tested concentrations from 200 to 500 mM (Figure S3). We started with 200 mM conditions because, in EPPS-based seedless gold nanostar synthesis, only short spikes were formed when the concentration was below 120 mM.³⁴ At this low concentration, fewer EPPS molecules were available to bind and direct growth, which led to low numbers of short spikes (Figure S4). Increasing the concentration from 200 to 400 mM, more gold ions were reduced onto specific facets of the central core surface, which resulted in higher numbers of longer spikes³⁵ as well as stronger LSP resonances at 700 and 900 nm (Figure S5a). At higher concentrations (500 mM), excess EPPS molecules accelerated the reduction of gold ions and promoted the nucleation and growth of undesired gold

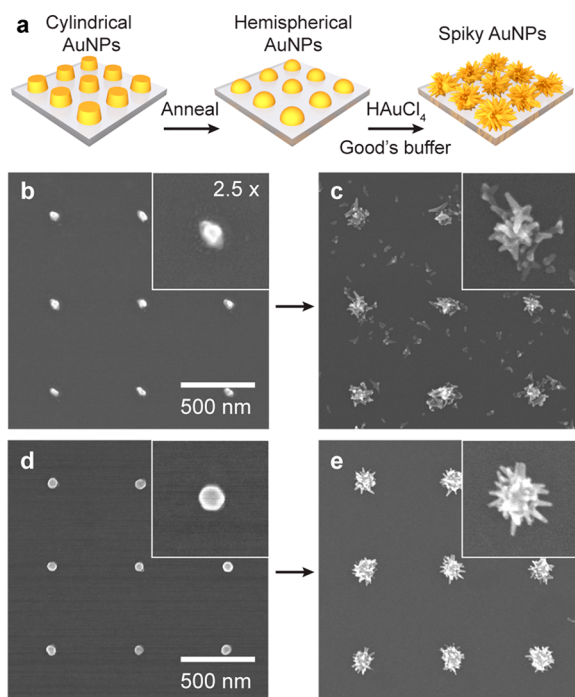


Figure 1. Scaffold-guided synthesis of spiky AuNPs with Good's buffer. (a) Fabrication scheme of the AuNP scaffolds and spike growth. SEM images of the patterned AuNPs before (b) and after (c) spike formation from as-fabricated NPs. Patterned AuNPs before (d) and after (e) spike formation from thermally annealed NPs. The spiky AuNPs shown in parts c and e were synthesized under the same growth condition with a growth solution consisting of HAuCl_4 (0.4 mM) and EPPS (400 mM) at pH 7.5.

assisted nanoscale embossing)²⁹ and PEEL (photolithography, etching, electron-beam metal deposition, and lift-off)³⁰ methods (Supporting Information Figure S1). We started with diameter $d = 70\text{ nm}$ for the patterned AuNPs based on the sizes of AuNP seeds in the EPPS buffer-based colloidal synthesis of spiky AuNPs.¹⁵ After SANE and PEEL, due to the nanoscale

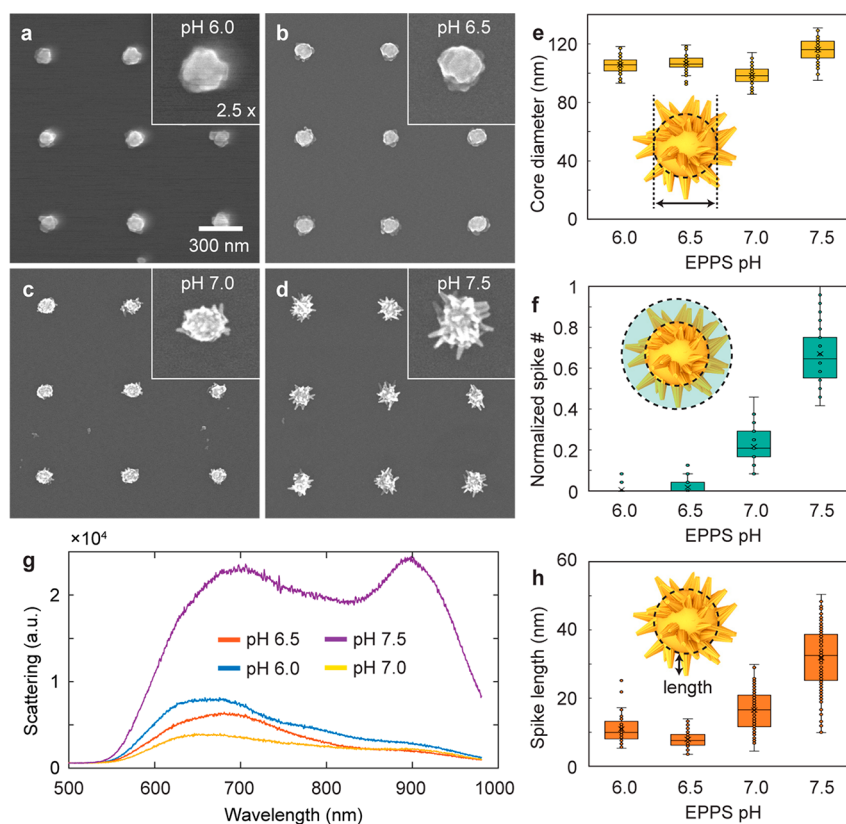


Figure 2. Spike growth on patterned AuNPs with different solution pH. SEM images of spiky AuNPs synthesized with pH values of (a) 6.0, (b) 6.5, (c) 7.0, and (d) 7.5. (e) Core diameter, (f) normalized peripheral spike number, (g) dark-field scattering spectra, and (h) estimated spike length. Top-view schemes of spiky AuNPs in parts e–g illustrate the parameters for morphology analysis. The growth solution of HAuCl₄ (0.4 mM) and EPPS (400 mM) was kept fixed for all spiky NP growth from $d = 70$ nm seeds.

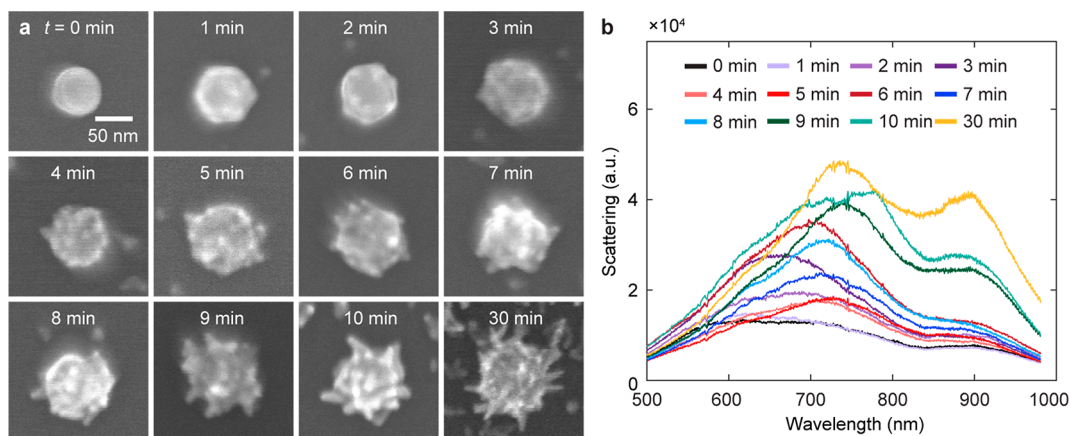


Figure 3. Reaction intermediates during the spike growth on the patterned AuNP scaffolds at the early stage. (a) SEM images and (b) dark-field scattering spectra of representative reaction intermediates of the spiky AuNPs at 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and 30 min. Three replicates each time were compared for consistency. The growth solution of HAuCl₄ (0.4 mM) and EPPS (400 mM) at pH 7.5 was kept fixed for all NP growth from $d = 70$ nm seeds. The samples were removed from the growth solution at the target time points, rinsed with deionized water, and dried with N₂ gas before measurements.

nanostars in the colloidal solution. The limited availability of gold ions for spike growth led to a decrease in spike number and ca. 10 nm reduction in spike length; hence, there was a slight blue shift of the longer-wavelength LSP resonance.

Since final NP morphologies depend on the crystallinity and shape of early and intermediate structures that direct the growth process,^{9,10} we characterized the growth mechanism of large spiky AuNPs by correlating their morphologies and scattering

spectra as a function of reaction time (Figure 3). We focused on conditions that produced the highest number of spikes (400 mM EPPS, 0.4 mM HAuCl₄ with pH 7.5). The initial seeds were hemispherical, with only a single LSP resonance at 615 nm ($t = 0$ min). After HAuCl₄ was injected in EPPS buffer ($t = 1–3$ min), there was an immediate red shift in the LSP wavelength with increased intensity because of the increase in the NP size and the formation of faceted features. After $t = 4$ min, small protrusions

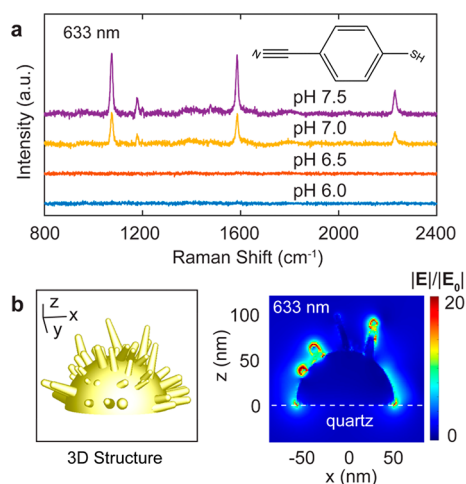


Figure 4. SERS properties of the patterned AuNPs synthesized with different solution pH. (a) Representative Raman spectra of 4-MBN monolayer on AuNPs synthesized with different solution pH. The Raman spectra were measured with a 633 nm excitation laser. (b) Spatial distribution of electromagnetic near-field ($|E|/|E_0|$) of a single spiky AuNP based on the morphology of AuNPs synthesized with EPPS (400 mM) at pH 7.5. The simulation used an excitation light with a wavelength of 633 nm. The color bar was set to 20 to provide better contrast.

started to develop, and the LSP resonance further red-shifted in wavelength (ca. 700 nm). At longer reaction times ($t = 5$ – 9 min), the NP size and the spike number and length continued to increase until there were two distinct peaks in the scattering spectra; this second LSP peak around 900 nm emerged as the spikes became longer. At $t = 30$ min, the highly branched AuNPs formed with high-aspect-ratio and sharp spikes, together with a significant increase in the second LSP resonance. Similar to previous work on seedless syntheses of gold nanostars using Good's buffers,⁹ the seed-mediated spike growth is also a two-step reaction.

Sharp tips (radius of curvature < 5 nm) can generate localized electromagnetic near-fields that result in strong SERS signals.³⁶ To test spiky AuNP arrays as SERS substrates, we functionalized the AuNPs with 4-mercaptobenzonitrile (4-MBN), a common Raman dye. Single-particle Raman measurements of 4-MBN-functionalized spiky AuNPs were performed on different particles in the array. The Raman spectra showed consistent intensities across the array (excitation at $\lambda = 633$ nm, near LSP resonances), which confirms the uniformity of the spike features to generate reproducible optical responses. Spiky AuNPs synthesized at pH 7.0 and 7.5 showed characteristic signatures of 4-MBN at 1075, 1177, 1586, and 2230 cm^{-1} (Figure 4a). The absence of Raman signals from AuNPs synthesized at solution pH 6.0 and 6.5 can be attributed to their lack of sharp tips, which are needed to generate enhanced electromagnetic near-fields. Spiky AuNPs with the highest spike density on larger surface area NP seeds showed the strongest SERS signals (Figure S5b).

Figure 4b represents the modeled nanostructure as a hemispherical core with protruding spikes based on the dimensions from Figure 2d. At the Raman excitation wavelength, the calculated near-field distribution showed that the strongest fields ($|E|/|E_0| = 36$) were concentrated near the tips of spikes with lower aspect ratios. The electromagnetic near-fields can be further enhanced by exciting plasmonic particles at their LSP wavelengths.³⁷ Since spiky AuNPs have two LSP resonances at 700 and 900 nm (Figure 2g), we also calculated the electromagnetic near-fields at these wavelengths (Figure S6). Increased enhancement ($|E|/|E_0| = 57$) was observed near the tips of lower-aspect-ratio spikes at 700 nm, and an even greater electromagnetic field intensity ($|E|/|E_0| = 175$) was observed near the tips of higher-aspect-ratio spikes at 900 nm.

To grow spikes on seed sizes larger than that easily realized in colloidal form, patterned AuNP scaffolds with $d = 100, 130, 160$, and 190 nm were used with the optimal growth solution for high numbers of long spikes (Figures 5a–d). After growth, the sizes of the central cores all increased (Figure S7a). Spiky AuNPs starting from initial seed diameters of 70 and 100 nm showed an increase of 45 nm; however, for spiky AuNPs prepared from

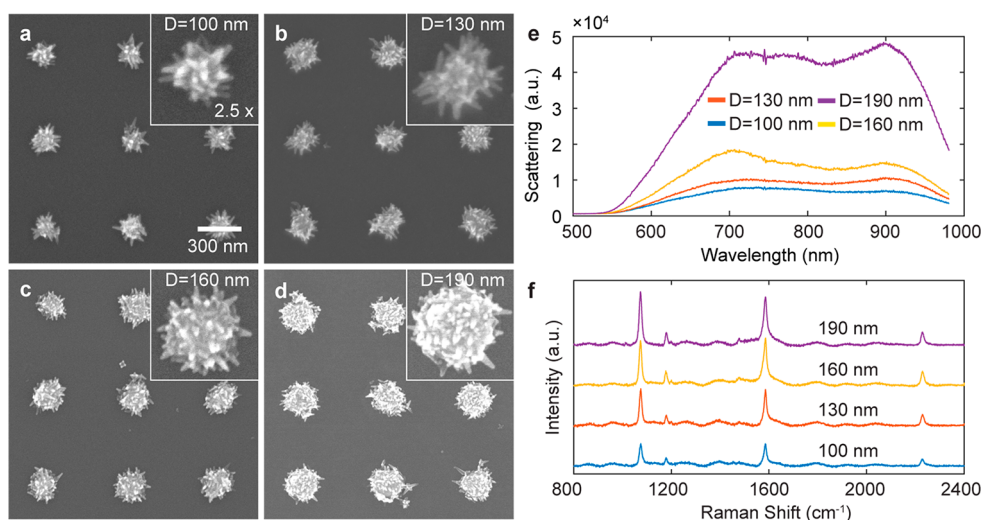


Figure 5. Growth of spikes on AuNP seeds larger than 100 nm. SEM images of spiky AuNPs synthesized with the initial seed diameters of (a) 100, (b) 130, (c) 160, and (d) 190 nm. (e) Dark-field scattering spectra of the spiky AuNPs. (f) Representative Raman spectra of 4-MBN-functionalized spiky AuNPs. The characteristic Raman peak at 1075 cm^{-1} , attributed to the C–S stretching mode of 4-MBN, exhibited intensities of 137, 228, 275, and 330 for spiky AuNPs with initial seed sizes of 100, 130, 160, and 190 nm, respectively. Larger spiky AuNPs had higher signal intensities because of a higher number of spikes, which resulted in an increased number of hot spots.

patterned seeds with $d = 130, 160,$ and 190 nm, the increase was less than 15 nm. Spikes were formed on all of the scaffolds, and spiky AuNPs showed two LSP resonances at wavelengths around 710 and 900 nm (Figure 5e). As the AuNP size increased, more spikes were grown (Figure S7b), which also increased the scattering and SERS signals (Figure 5f). The shorter spikes of the larger spiky AuNPs (Figure S7c) led to a slight blue shift of the second LSP resonance, in agreement with longer spikes having a red-shifted LSP resonance.¹⁵ Top-down fabrication methods to pattern scaffolds can easily produce structures over large areas (cm^2) (Figure S8).

In summary, we demonstrated that combining nanofabricated scaffolds with solution-based synthesis can create spiky AuNPs with tunable sizes, shapes, and optical properties. The use of thermally annealed AuNPs as large seeds enables the growth of spiky AuNPs with a high degree of structural uniformity. By correlating the structural and optical properties of AuNPs at specific time points, we determined key intermediate species during the spike growth process and provided insight into the mechanism of anisotropic NP synthesis. In addition, we established a set of parameters to manipulate the spike number and dimensions for large seed sizes (>100 nm). Large spiky AuNPs that exhibit numerous locations of strong electromagnetic fields will be beneficial in diverse plasmonic applications, from single-particle sensing to localized light harvesting and hot-electron-mediated energy conversion.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.3c03778>.

Fabrication procedures of the patterned AuNPs; SEM images, morphology analysis, and optical properties of spiky AuNPs synthesized with different EPPS concentrations; near-field enhancement of a single spiky AuNP at the LSP wavelength; morphology analysis of spiky AuNPs synthesized with different initial seed sizes; SEM images of large-area spiky AuNP arrays (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Teri W. Odom — Department of Chemistry and Department of Materials Science and Engineering, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0002-8490-292X; Email: todom@northwestern.edu

Authors

Jin Jia — Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0002-2194-5607

Nadia Metzkwow — Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; Present Address: Physical Chemistry, Technical University Dresden, Dresden 01069, Germany

Sang-Min Park — Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0002-4306-0375

Yuhao Leo Wu — Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0009-0002-6227-3963

Alexander D. Sample — Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0002-4561-8619

Bundit Diloknawarit — Department of Materials Science and Engineering, Northwestern University, Evanston, Illinois 60208, United States

Insub Jung — Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0003-0827-9280

Complete contact information is available at: <https://pubs.acs.org/10.1021/acs.nanolett.3c03778>

Author Contributions

J.J. and N.M. contributed equally to this work. J.J., N.M., and T.W.O. conceived the idea of growing spikes on substrate-bound AuNPs. J.J., N.M., Y.L.W., and A.D.S. fabricated and optimized the patterned AuNPs. J.J., N.M., Y.L.W., and I.J. grew spikes on the patterned AuNPs. J.J. and N.M. characterized the morphology of the spiky AuNPs. S.-M.P. performed the FDTD numerical simulation. B.D. carried out the dark-field scattering measurements. T.W.O. guided the experimental and theoretical investigations. J.J., N.M., and T.W.O. analyzed the data and wrote the manuscript. All authors commented on and revised the manuscript.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors thank Dr. Priscilla Choo for the help during the initial trial experiments. This work was supported by the National Science Foundation (NSF) (CHE-2305039, Y.L.W., B.D., and T.W.O.), the Vannevar Bush Faculty Fellowship from the U.S. Department of Defense (DOD) (N00014-17-1-3024, J.J. and T.W.O.), and the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) (GRK 2767 — Project No. 451785257, N.M.). This work made use of the EPIC, SPID, and NUFAB facilities of Northwestern University's NUANCE Center, which has received support from the SHyNE Resource (NSF ECCS-2025633), the International Institute of Nanotechnology (IIN) at Northwestern, and Northwestern's MRSEC program (NSF DMR-2308691). This research was supported in part through the computational resources and staff contributions provided for the Quest high performance computing facility at Northwestern University, which is jointly supported by the Office of the Provost, the Office for Research, and Northwestern University Information Technology.

■ REFERENCES

- (1) Murphy, C. J.; Sau, T. K.; Gole, A. M.; Orendorff, C. J.; Gao, J.; Gou, L.; Hunyadi, S. E.; Li, T. Anisotropic Metal Nanoparticles: Synthesis, Assembly, and Optical Applications. *J. Phys. Chem. B* **2005**, *109* (29), 13857–13870.
- (2) Lu, X.; Rycenga, M.; Skrabalak, S. E.; Wiley, B.; Xia, Y. Chemical Synthesis of Novel Plasmonic Nanoparticles. *Annu. Rev. Phys. Chem.* **2009**, *60* (1), 167–192.
- (3) Ni, B.; Mychinko, M.; Gómez-Graña, S.; Morales-Vidal, J.; Obelleiro-Liz, M.; Heyvaert, W.; Vila-Liarte, D.; Zhuo, X.; Albrecht, W.; Zheng, G.; González-Rubio, G.; Taboada, J. M.; Obelleiro, F.; López, N.; Pérez-Juste, J.; Pastoriza-Santos, I.; Cölfen, H.; Bals, S.; Liz-Marzán, L. M. Chiral Seeded Growth of Gold Nanorods Into Fourfold Twisted Nanoparticles with Plasmonic Optical Activity. *Adv. Mater.* **2023**, *35* (1), No. 2208299.

- (4) Obelleiro-Liz, M.; Martín, V. F.; Solís, D. M.; Taboada, J. M.; Obelleiro, F.; Liz-Marzán, L. M. Influence of Geometrical Parameters on the Optical Activity of Chiral Gold Nanorods. *Adv. Opt. Mater.* **2023**, *11*, No. 2203090.
- (5) Kelly, K. L.; Coronado, E.; Zhao, L. L.; Schatz, G. C. The Optical Properties of Metal Nanoparticles: The Influence of Size, Shape, and Dielectric Environment. *J. Phys. Chem. B* **2003**, *107* (3), 668–677.
- (6) Yuan, H.; Khoury, C. G.; Hwang, H.; Wilson, C. M.; Grant, G. A.; Vo-Dinh, T. Gold Nanostars: Surfactant-free Synthesis, 3D modelling, and Two-photon Photoluminescence Imaging. *Nanotechnology* **2012**, *23* (7), No. 075102.
- (7) Coughlin, E. E.; Hu, J.; Lee, A.; Odom, T. W. Light-Mediated Directed Placement of Different DNA Sequences on Single Gold Nanoparticles. *J. Am. Chem. Soc.* **2021**, *143* (10), 3671–3676.
- (8) Dardir, K.; Wang, H.; Martin, B. E.; Atzampou, M.; Brooke, C. B.; Fabris, L. SERS Nanoprobe for Intracellular Monitoring of Viral Mutations. *J. Phys. Chem. C* **2020**, *124* (5), 3211–3217.
- (9) Chandra, K.; Rugg, B. K.; Ratner, M. A.; Wasielewski, M. R.; Odom, T. W. Detecting and Visualizing Reaction Intermediates of Anisotropic Nanoparticle Growth. *J. Am. Chem. Soc.* **2018**, *140* (9), 3219–3222.
- (10) Choo, P.; Arenas-Esteban, D.; Jung, I.; Chang, W. J.; Weiss, E. A.; Bals, S.; Odom, T. W. Investigating Reaction Intermediates during the Seedless Growth of Gold Nanostars Using Electron Tomography. *ACS Nano* **2022**, *16* (3), 4408–4414.
- (11) Chandra, K.; Kumar, V.; Werner, S. E.; Odom, T. W. Separation of Stabilized MOPS Gold Nanostars by Density Gradient Centrifugation. *ACS Omega* **2017**, *2* (8), 4878–4884.
- (12) De Silva Indrasekara, A. S.; Johnson, S. F.; Odion, R. A.; Vo-Dinh, T. Manipulation of the Geometry and Modulation of the Optical Response of Surfactant-Free Gold Nanostars: A Systematic Bottom-Up Synthesis. *ACS Omega* **2018**, *3* (2), 2202–2210.
- (13) Rodríguez-Lorenzo, L.; Romo-Herrera, J. M.; Pérez-Juste, J.; Alvarez-Puebla, R. A.; Liz-Marzán, L. M. Reshaping and LSPR Tuning of Au Nanostars in the Presence of CTAB. *J. Mater. Chem.* **2011**, *21* (31), 11544–11549.
- (14) Barbosa, S.; Agrawal, A.; Rodríguez-Lorenzo, L.; Pastoriza-Santos, I.; Alvarez-Puebla, R. A.; Kornowski, A.; Weller, H.; Liz-Marzán, L. M. Tuning Size and Sensing Properties in Colloidal Gold Nanostars. *Langmuir* **2010**, *26* (18), 14943–14950.
- (15) Pallares, R. M.; Stilson, T.; Choo, P.; Hu, J.; Odom, T. W. Using Good's Buffers To Control the Anisotropic Structure and Optical Properties of Spiky Gold Nanoparticles for Refractive Index Sensing. *ACS Appl. Nano Mater.* **2019**, *2* (8), 5266–5271.
- (16) Odom, T. W.; Nehl, C. L. How Gold Nanoparticles Have Stayed in the Light: The 3M's Principle. *ACS Nano* **2008**, *2* (4), 612–616.
- (17) Nguyen, Q. N.; Wang, C.; Shang, Y.; Janssen, A.; Xia, Y. Colloidal Synthesis of Metal Nanocrystals: From Asymmetrical Growth to Symmetry Breaking. *Chem. Rev.* **2023**, *123* (7), 3693–3760.
- (18) Modaresialam, M.; Chehadi, Z.; Bottein, T.; Abbarchi, M.; Grosso, D. Nanoimprint Lithography Processing of Inorganic-Based Materials. *Chem. Mater.* **2021**, *33* (14), 5464–5482.
- (19) Chen, K.; Rajeeva, B. B.; Wu, Z.; Rukavina, M.; Dao, T. D.; Ishii, S.; Aono, M.; Nagao, T.; Zheng, Y. Moiré Nanosphere Lithography. *ACS Nano* **2015**, *9* (6), 6031–6040.
- (20) Burek, M. J.; Greer, J. R. Fabrication and Microstructure Control of Nanoscale Mechanical Testing Specimens via Electron Beam Lithography and Electroplating. *Nano Lett.* **2010**, *10* (1), 69–76.
- (21) Gates, B. D.; Xu, Q.; Stewart, M.; Ryan, D.; Willson, C. G.; Whitesides, G. M. New Approaches to Nanofabrication: Molding, Printing, and Other Techniques. *Chem. Rev.* **2005**, *105* (4), 1171–1196.
- (22) Abid, N.; Khan, A. M.; Shujait, S.; Chaudhary, K.; Ikram, M.; Imran, M.; Haider, J.; Khan, M.; Khan, Q.; Maqbool, M. Synthesis of Nanomaterials using Various Top-down and Bottom-up Approaches, Influencing Factors, Advantages, and Disadvantages: A Review. *Adv. Colloid Interface Sci.* **2022**, *300*, No. 102597.
- (23) Liddle, J. A.; Gallatin, G. M. Nanomanufacturing: A Perspective. *ACS Nano* **2016**, *10* (3), 2995–3014.
- (24) Wang, D.; Wang, W.; Knudson, M. P.; Schatz, G. C.; Odom, T. W. Structural Engineering in Plasmon Nanolasers. *Chem. Rev.* **2018**, *118* (6), 2865–2881.
- (25) Guan, J.; Park, J.-E.; Deng, S.; Tan, M. J. H.; Hu, J.; Odom, T. W. Light–Matter Interactions in Hybrid Material Metasurfaces. *Chem. Rev.* **2022**, *122* (19), 15177–15203.
- (26) Deng, S.; Li, R.; Park, J.-E.; Guan, J.; Choo, P.; Hu, J.; Smeets, P. J. M.; Odom, T. W. Ultranarrow Plasmon Resonances from Annealed Nanoparticle Lattices. *Proc. Natl. Acad. Sci. U. S. A.* **2020**, *117* (38), 23380–23384.
- (27) Golze, S. D.; Hughes, R. A.; Rouvimov, S.; Neal, R. D.; Demille, T. B.; Neretina, S. Plasmon-Mediated Synthesis of Periodic Arrays of Gold Nanoplates Using Substrate-Immobilized Seeds Lined with Planar Defects. *Nano Lett.* **2019**, *19* (8), 5653–5660.
- (28) Demille, T. B.; Hughes, R. A.; Dominique, N.; Olson, J. E.; Rouvimov, S.; Camden, J. P.; Neretina, S. Large-area Periodic Arrays of Gold Nanostars Derived from HEPES-, DMF-, and Ascorbic-acid-driven Syntheses. *Nanoscale* **2020**, *12* (31), 16489–16500.
- (29) Lee, M. H.; Huntington, M. D.; Zhou, W.; Yang, J.-C.; Odom, T. W. Programmable Soft Lithography: Solvent-Assisted Nanoscale Embossing. *Nano Lett.* **2011**, *11* (2), 311–315.
- (30) Henzie, J.; Lee, M. H.; Odom, T. W. Multiscale Patterning of Plasmonic Metamaterials. *Nat. Nanotechnol.* **2007**, *2* (9), 549–554.
- (31) Vinnacombe-Willson, G. A.; Conti, Y.; Stefancu, A.; Weiss, P. S.; Cortés, E.; Scarabelli, L. Direct Bottom-Up In Situ Growth: A Paradigm Shift for Studies in Wet-Chemical Synthesis of Gold Nanoparticles. *Chem. Rev.* **2023**, *123* (13), 8488–8529.
- (32) Zare, I.; Yarak, M. T.; Speranza, G.; Najafabadi, A. H.; Shourangiz-Haghighi, A.; Nik, A. B.; Manshian, B. B.; Saraiva, C.; Soenen, S. J.; Kogan, M. J.; Lee, J. W.; Apollo, N. V.; Bernardino, L.; Araya, E.; Mayer, D.; Mao, G.; Hamblin, M. R. Gold Nanostructures: Synthesis, Properties, and Neurological Applications. *Chem. Soc. Rev.* **2022**, *51* (7), 2601–2680.
- (33) Xia, Y.; Xia, X.; Peng, H.-C. Shape-Controlled Synthesis of Colloidal Metal Nanocrystals: Thermodynamic versus Kinetic Products. *J. Am. Chem. Soc.* **2015**, *137* (25), 7947–7966.
- (34) Chandra, K.; Culver, K. S. B.; Werner, S. E.; Lee, R. C.; Odom, T. W. Manipulating the Anisotropic Structure of Gold Nanostars using Good's Buffers. *Chem. Mater.* **2016**, *28* (18), 6763–6769.
- (35) Webb, J. A.; Erwin, W. R.; Zarick, H. F.; Aufrecht, J.; Manning, H. W.; Lang, M. J.; Pint, C. L.; Bardhan, R. Geometry-Dependent Plasmonic Tunability and Photothermal Characteristics of Multi-branched Gold Nanoantennas. *J. Phys. Chem. C* **2014**, *118* (7), 3696–3707.
- (36) Yuan, L.; Lou, M.; Clark, B. D.; Lou, M.; Zhou, L.; Tian, S.; Jacobson, C. R.; Nordlander, P.; Halas, N. J. Morphology-Dependent Reactivity of a Plasmonic Photocatalyst. *ACS Nano* **2020**, *14* (9), 12054–12063.
- (37) Atta, S.; Tsoulos, T. V.; Fabris, L. Shaping Gold Nanostar Electric Fields for Surface-Enhanced Raman Spectroscopy Enhancement via Silica Coating and Selective Etching. *J. Phys. Chem. C* **2016**, *120* (37), 20749–20758.